

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

EE-703(B)-CBGS

B.Tech., VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Soft Computing Techniques and Application

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Define Soft Computing. Mention the applications of soft computing. 7
b) With graphical representations, explain the activation functions used in Artificial Neural Network (ANN). 7
2. a) What is meant by genetic algorithm? With a neat flowchart, explain the operation of a simple genetic algorithm. 7
b) Write various advantages and disadvantages of Genetic algorithm. 7
3. a) Draw the architecture of Back-Propagation network. Write its testing algorithm. 7
b) Explain four mutation methods? 7
4. a) Explain back propagation algorithm with the help of a block diagram and a suitable example. 7
b) What are the classical notations in statistics, linear algebra and machine learning? and what are the connections between these notations? 7

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[2]

- 5. a) Define statistical theory and how it is performed in Soft Computing? 7
- b) Explain the concept of Bayesian theorem with an example. 7
- 6. a) Define Bayesian learning and how it impacts in Soft Computing? 7
- b) Distinguish between evolution strategies and genetic algorithms. 7
- 7. a) Define Linear System and a nonlinear system. 7
- b) What is a hybrid intelligent control? Explain. 7
- 8. Explain the following terms with example: 14
 - i) Sampling distribution
 - ii) RBF Network
 - iii Coefficient of variation
 - iv) Multi member Evolution Strategies (ES)
